

A Taxonomy of Geometric Space for Diffusion Models



Prithika Narayanan, Daria Godorozha, Mateusz Mikolajczak, Aditya Shethwala

Diffusion models are still largely derived on a space by space basis, with the underlying geometry determining their construction. Symmetry supplies the stationary measure, and finite volume decides whether it can be kept without a drift. Five geometric axes reduce to one isometry group and three dichotomies, with curvature and topology entering the qualitative prescription only through compactness. The isometry group also fixes the kernel basis, while boundaries and discreteness determine whether the dynamics require a reflection or a discrete generator. This framework yields a lookup table spanning eleven spaces, along with an explicit diffusion model on hyperbolic space whose prior is normalisable for any positive drift, and which mixes at a rate the curvature sets.

Table 1: Geometric profile. What each space is: curvature, compactness, boundary, topology, and the symmetry group that organises the rest. Row order matches Table 2.

Read the two tables together: locate a space by its profile in Table 1, then read its prescription across in Table 2.

Space	Curvature	Compact?	Boundary	Topology	Symmetry group
Euclidean space $\mathbb{R}^n$	Flat ($K = 0$)	No	None	Contractible; π_1 trivial	$E(n) = O(n) \ltimes \mathbb{R}^n$ (flat homogeneous)
n -sphere $S^n \subset \mathbb{R}^{n+1}$	Positive ($K = +1$)	Yes	None	Simply connected ($n \geq 2$); $\pi_1(S^1) = \mathbb{Z}$	$O(n+1)$; $S^n \cong O(n+1)/O(n)$
Flat torus $T^n = \mathbb{R}^n/\mathbb{Z}^n$	Flat ($K = 0$)	Yes	None	$\pi_1 = \mathbb{Z}^n$ (not simply connected)	Translation subgroup T^n
Rotations $\text{SO}(3)$ (compact Lie group)	Non-negative (bi-invariant)	Yes	None	$\pi_1 = \mathbb{Z}_2$; $\text{SO}(3) \cong \mathbb{RP}^3$ (cover $\text{SU}(2)$)	Left/right translation; bi-invariant metric
Rigid motions $\text{SE}(3) = \text{SO}(3) \ltimes \mathbb{R}^3$	Mixed; no bi-invariant metric	No	None	$\pi_1 = \mathbb{Z}_2$; $\mathbb{R}^3$ contractible	Left-invariant; semidirect product
Hyperbolic space $\mathbb{H}^n$	Negative ($K = -1$)	No	None (∂_∞ not a manifold boundary)	Contractible (Hadamard $\cong \mathbb{R}^n$)	$O^+(n, 1)$; $\mathbb{H}^n \cong \text{SO}^+(n, 1)/\text{SO}(n)$
SPD cone $\text{Sym}^+(n)$ (affine-invariant metric)	Non-positive (Hadamard/CAT(0))	No	None intrinsically (singular PSD at ∞)	Contractible (open convex cone)	$\text{GL}(n)$ congruence $P \mapsto APA^\top$; $\cong \text{GL}(n)/O(n)$
Probability simplex Δ^{n-1}	Positive (Fisher–Rao)	Yes	Yes — faces where some $x_i = 0$	Contractible (convex); faces form boundary	S_n permutations; $\sqrt{\cdot}$ -isometry to S^{n-1} orthant
Stiefel $V_k(\mathbb{R}^n)$; Grassmann $\text{Gr}(k, n)$	Non-negative (normal homogeneous)	Yes	None	Non-trivial (e.g. $\pi_1(\text{Gr}) = \mathbb{Z}_2$, real)	$O(n)$ acts transitively; homogeneous
Graphs/categorical discrete data	N/A (not a smooth manifold)	N/A (discrete)	N/A	Combinatorial; no differential structure	S_n (node permutations/automorphisms)
Stratified/BHV tree space $\mathcal{T}_n$	CAT(0) but singular	No	Strata (collapsed edges)	Cone-complex; not a manifold	Label permutations; local orthant charts

Table 2: Diffusion prescription. What geometry forces on the process: admissible forward dynamics, the stationary prior, the obstruction and its remedy, the natural kernel basis, and representative methods. Row order matches Table 1.

Space	Forward process	Stationary prior	Obstruction & remedy	Kernel/eigenbasis	Representative methods
Euclidean space $\mathbb{R}^n$	VP/OU SDE (mean-reverting); or VE (variance-exploding)	Isotropic Gaussian $\mathcal{N}(0, I)$ (VP; approximate for VE)	Infinite volume $\rightarrow$ drift needed for a normalisable prior	Gaussian heat kernel; Hermite/Fourier	DDPM; Score-SDE
n -sphere S^n	Intrinsic heat flow/Riemannian BM (Laplace–Beltrami)	Uniform (surface measure)	None (compact) — no drift required	Spherical harmonics $Y_{\ell m}$	Riemannian SGM; directional data
Flat torus T^n	Wrapped BM (heat kernel = wrapped Gaussian)	Uniform on T^n	None (compact); must respect periodicity	Fourier modes $e^{ik \cdot x}$, $k \in \mathbb{Z}^n$	Torsional diffusion (conformers)
Rotations $\text{SO}(3)$	Group BM = IGSO(3) (its heat kernel)	Haar measure (= uniform)	None (compact group) — no drift	Wigner- D functions D_{mn}^ℓ	FrameDiff; RFDiffusion
Rigid motions $\text{SE}(3)$	Factorised: BM on $\text{SO}(3) \times \text{OU}$ on $\mathbb{R}^3$	Haar($\text{SO}(3)$) $\times \mathcal{N}(\mathbb{R}^3)$	Infinite volume in translation $\rightarrow$ drift; rotation needs none	Wigner- $D \times$ Gaussian	$\text{SE}(3)$ diffusion (poses, docking, robotics)
Hyperbolic space $\mathbb{H}^n$	Riemannian BM (transient) + drift; or wrapped normal	Hyperbolic Gaussian $\mu_c \propto e^{-\frac{c}{2}r^2}$; or wrapped normal	Infinite volume, $K < 0 \rightarrow$ transience; drift required	Closed-form heat kernel (McKean)	Under-served; hierarchical embeddings (open)
SPD cone $\text{Sym}^+(n)$	Affine-invariant BM; log-Euclidean/Wishart-type	Log-normal/Wishart-type (via drift)	Infinite volume $\rightarrow$ drift; iterates must stay PD	Zonal/matrix spherical functions	Covariance; DTI tensors; connectivity
Probability simplex Δ^{n-1}	Wright–Fisher/Jacobi; or $\sqrt{\cdot}$ -map to sphere; or reflected BM	Dirichlet($1, \dots, 1$) = uniform	Boundary $\rightarrow$ reflection or a face-preserving process	Jacobi polynomials; harmonics via $\sqrt{\cdot}$ -map	Compositional data; Dirichlet diffusion
Stiefel/Grassmann	Riemannian BM on the homogeneous space	Uniform ($O(n)$ -invariant measure)	None (compact); retract to enforce constraint	Zonal/representation-theoretic functions	Frames, subspaces, PCA-type constraints
Graphs/discrete	CTMC noising (uniform/absorbing/marginal) — not an SDE	Uniform over categories, or δ at absorbing [MASK]	Discrete $\rightarrow$ Markov jump; impose permutation equivariance	Transition-matrix exponential	D3PM; DiGress
Stratified/tree space	No canonical BM across strata (candidates: sticky/reflected)	Open — no agreed invariant law	Singular strata $\rightarrow$ manifold diffusion undefined	—	Frontier/open problem

The method column is a consequence rather than a coincidence. The isotropic Gaussian on $\text{SO}(3)$ in RFDiffusion is the heat kernel relaxing to Haar measure, and the Gaussian prior of DDPM is what an Ornstein–Uhlenbeck drift relaxes to once infinite volume rules the undrifted process out. Each row is the same statement, that symmetry fixes the prior, specialised to a different space.

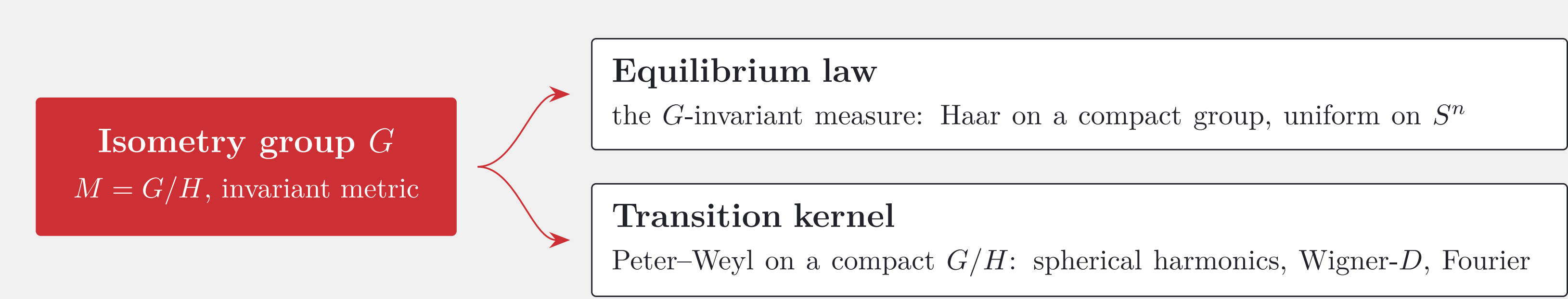
What the Taxonomy Is For

A design tool. Confronted with an unfamiliar data space, a practitioner must answer four coupled questions at once: what the stationary prior should be, whether the forward process needs a drift, whether it must respect a boundary, and which basis yields closed form transition kernels. Once the isometry group and the three dichotomies are known, the admissible construction is essentially determined. The table records that determination for eleven representative spaces, so the answer can be read off directly.

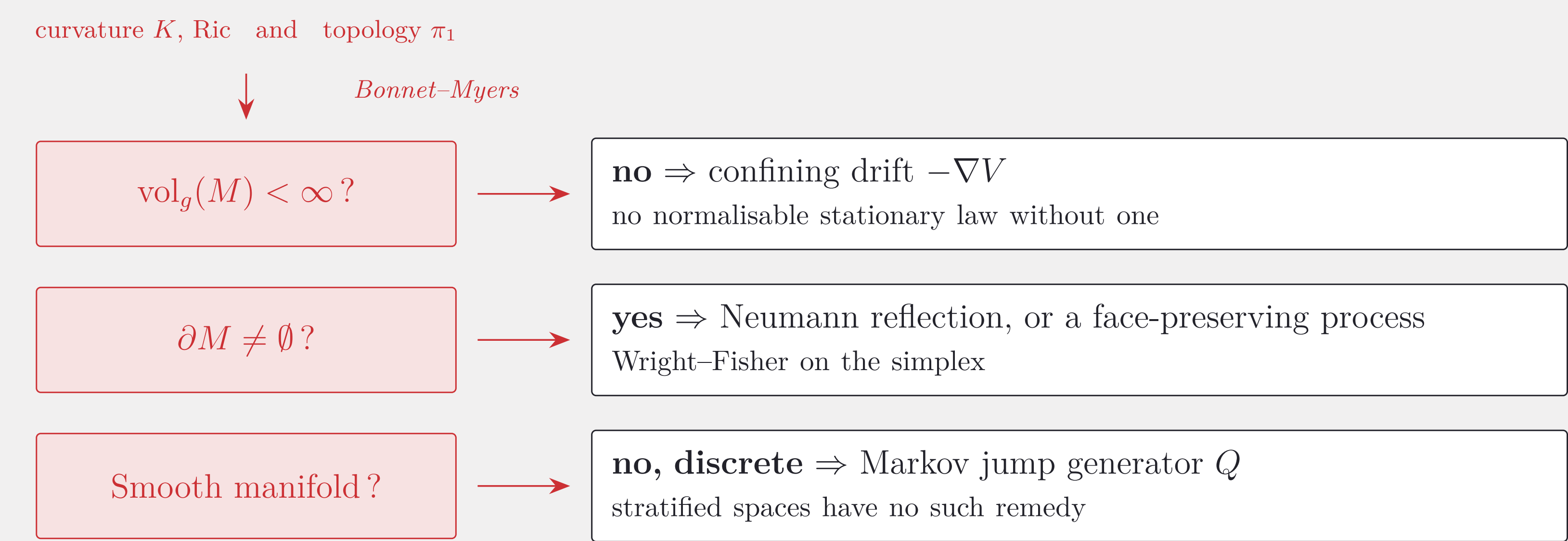
An organising principle. The same claim unifies methods that appear unrelated. In DDPM and variance-preserving SDEs the Gaussian prior is the stationary law of an Ornstein–Uhlenbeck drift, required only because $\mathbb{R}^n$ carries infinite volume. The isotropic Gaussian on $\text{SO}(3)$ used in FrameDiff and RFDiffusion is instead the heat kernel of undrifted Brownian motion, whose stationary law is Haar measure because the group is compact. Discrete graph diffusion abandons the second order generator altogether for a Markov rate matrix. Each is a single statement, that symmetry fixes the prior, specialised to a different space. What separates them is simply which dichotomies each space trips.

Five geometric axes reduce to one isometry group and three dichotomies

(a) Invariance under the isometry group



(b) Three dichotomies fixing the generator and its domain



Curvature and topology are not independent axes: a positive Ricci lower bound forces finite volume (Bonnet–Myers), and both therefore act on the qualitative prescription only through the first condition. Curvature returns quantitatively, setting the confinement at which mixing is guaranteed.

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The Invariant Measure and the Drift

Corollary. Suppose $M = G/H$ carries a G -invariant Riemannian metric.

- (a) vol_g is G -invariant, and is the unique G -invariant measure up to positive scale; it is invariant for Brownian motion.
- (b) If G is compact then so is M , the normalised G -invariant measure is the unique stationary law, and no drift is needed. For $M = G$ a compact Lie group with a bi-invariant metric, that measure is Haar measure.
- (c) If M is non-compact with $\text{vol}_g(M) = \infty$, the G -invariant measure is not normalisable, and a prior requires a drift $-\nabla V$. The resulting μ_V is invariant only under $\{\gamma \in G : V \circ \gamma = V\}$, and the drift therefore breaks full G -invariance.

Part (c) holds on every space of infinite volume, $\mathbb{R}^n$ and $\mathbb{H}^n$ included. The prior is invariant under the isometries preserving V , not under all of G .

A Hyperbolic Diffusion with a Curvature-Set Rate

Theorem (a hyperbolic Gaussian with a curvature-set gap). On $\mathbb{H}^n$ ($n \geq 2$), fix a base point o , write $r = d(o, \cdot)$, and take the generator

$$\mathcal{L}_c = \Delta - \nabla W_c \cdot \nabla, \quad W_c = \tfrac{c}{2} r^2, \quad c > 0.$$

- (i) $d\mu_c \propto e^{-\frac{c}{2}r^2} d\text{vol}$ is a probability measure and the unique invariant law of the diffusion;
- (ii) if $c > n - 1$, then $\text{Ric} + \nabla^2 W_c \geq (c - (n - 1))g$, so $\text{CD}(c - (n - 1), \infty)$ holds and

$$\|P_t f - \mu_c(f)\|_{L^2(\mu_c)} \leq e^{-(c - (n - 1))t} \|f - \mu_c(f)\|_{L^2(\mu_c)}.$$

Part (i) gives a normalisable prior for every $c > 0$. The threshold in part (ii) is exactly the Ricci bound $\text{Ric} = -(n - 1)g$ on $\mathbb{H}^n$: curvature sets the confinement at which the gap is guaranteed, not whether a drift is needed. At constant curvature $-a^2$ it becomes $c > (n - 1)a^2$, recovering the Euclidean Ornstein–Uhlenbeck case as $a \rightarrow 0$.

Finite Volume, Not Compactness, Removes the Drift

Proposition (drift depends on volume; the rate is a separate, spectral matter). Let (M, g) be connected, complete and without boundary, with generator Δ .

- (a) If $\text{vol}_g(M) < \infty$, whether or not M is compact, the normalised volume is the unique invariant probability measure of the diffusion, with no drift.
- (b) Convergence is exponential at rate λ_1 whenever $\lambda_1 > 0$, where λ_1 is the bottom of the nonzero spectrum of $-\Delta$. A gap is guaranteed by compactness, through discreteness of the spectrum, but not by finite volume alone.

The pivot of the dichotomy is therefore finite against infinite volume, not compactness. A non-compact hyperbolic quotient of finite volume $\Gamma \backslash \mathbb{H}^n$ still has $\lambda_1 > 0$, yet $-\Delta$ acquires continuous spectrum $[(\frac{n-1}{2})^2, \infty)$, and λ_1 can be made arbitrarily small across such quotients. Compactness is a convenient sufficient condition for both a normalisable prior and a controlled rate, never a necessary one.

Limitations and Open Directions

The table shows where the entries remain incomplete. Each item below is an open direction rather than a gap in coverage.

- The SPD cone.* Diffusion on $\text{Sym}^+(n)$ can be constructed through the affine-invariant metric. The geometry singles out no stationary prior, and the choice between log-Euclidean and Wishart type dynamics remains unsettled.

- Tree space.* The Billera–Holmes–Vogtmann space of phylogenetic trees is a cone complex, and it lacks the smooth Riemannian structure that the Laplace–Beltrami generator above requires. Defining a well posed forward process where its orthants meet is open.

- Hyperbolic space.* The construction above settles the prior and the rate. Sharpening the constants, choosing a score parameterisation and discretisation, and validating on hierarchical data remain.

Scope is deliberately finite dimensional and metric, with a single invariant metric fixed per space. Alternative choices, log-Euclidean against affine-invariant on the SPD cone among them, can change which dynamics the geometry permits and are not compared here.